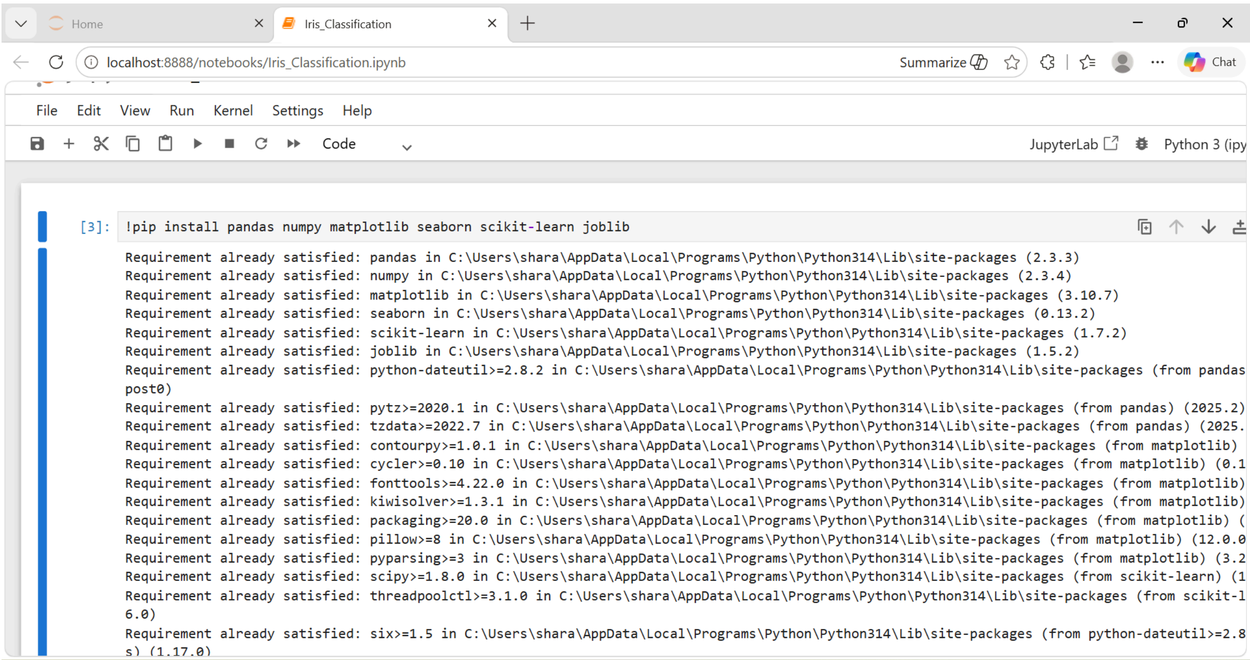
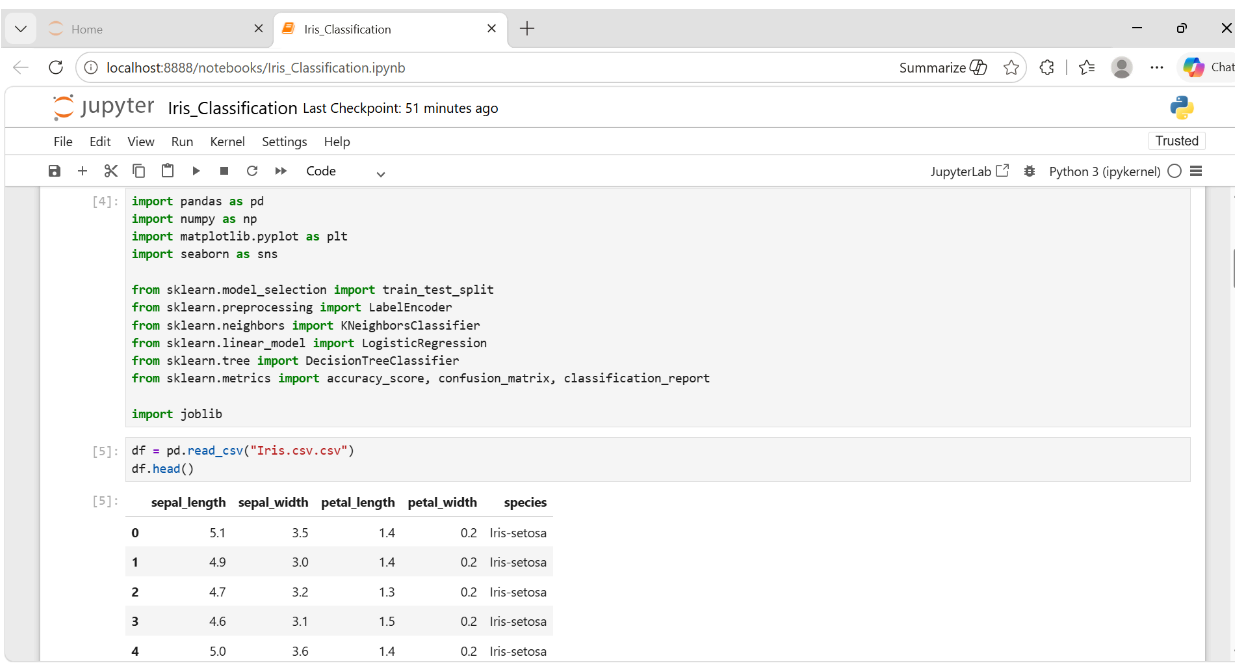
1. Installl Libraries



```
[3]: !pip install pandas numpy matplotlib seaborn scikit-learn joblib

Requirement already satisfied: pandas in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (2.3.3)
Requirement already satisfied: numpy in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (2.3.4)
Requirement already satisfied: matplotlib in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (3.10.7)
Requirement already satisfied: seaborn in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (0.13.2)
Requirement already satisfied: scikit-learn in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (1.7.2)
Requirement already satisfied: joblib in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (1.5.2)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas
post0)
Requirement already satisfied: pytz>=2020.1 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas) (2025.
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib)
Requirement already satisfied: cycler>=0.10 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib) (0.1
Requirement already satisfied: fonttools>=4.22.0 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib)
Requirement already satisfied: packaging>=20.0 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib) (
Requirement already satisfied: pillow>=8 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib) (12.0.0
Requirement already satisfied: pyparsing>=3 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib) (3.2
Requirement already satisfied: scipy>=1.8.0 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from scikit-learn) (1
Requirement already satisfied: threadpoolctl>=3.1.0 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from scikit-l
6.0)
Requirement already satisfied: six>=1.5 in C:\Users\shara\AppData\Local\Programs\Python\Python314\Lib\site-packages (from python-dateutil>=2.8
s) (1.17.0)
```



```
[4]: import pandas as pd
     import numpy as np
     import matplotlib.pyplot as plt
     import seaborn as sns

     from sklearn.model_selection import train_test_split
     from sklearn.preprocessing import LabelEncoder
     from sklearn.neighbors import KNeighborsClassifier
     from sklearn.linear_model import LogisticRegression
     from sklearn.tree import DecisionTreeClassifier
     from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

     import joblib

[5]: df = pd.read_csv("Iris.csv.csv")
     df.head()
```

[5]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

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Jupyter Iris_Classification Last Checkpoint: 52 minutes ago

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Code JupyterLab Python 3 (ipykernel)

[6]:
```
print(df.columns)
```
```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')
```

[7]:
```
df.info()
df.describe()
df.isnull().sum()
```
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

[7]:
```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

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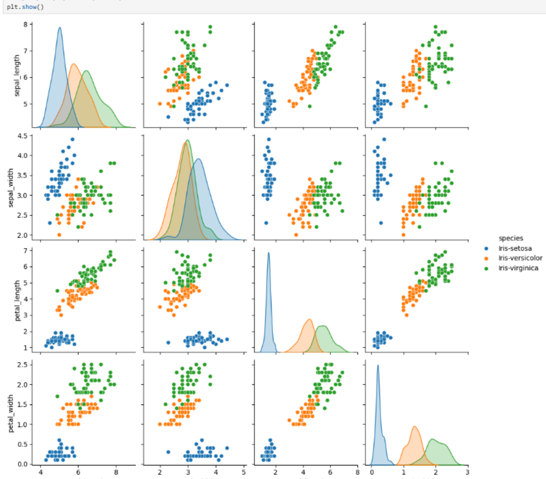
[25]:
```
sns.pairplot(df, hue='species')
plt.show()
```



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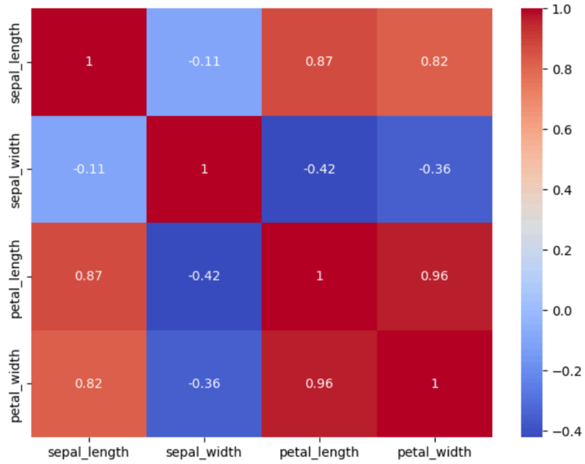
sepal_length sepal_width petal_length petal_width

[9]:
```python
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()
```



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sepal_length sepal_width petal_length petal_width

[14]:
```python
print(df.columns)
```

```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')
```

[16]:
```python
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

# Features and target
X = df.drop('species', axis=1)
y = df['species']

# Encode target labels
encoder = LabelEncoder()
y = encoder.fit_transform(y)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

[17]:
```python
knn = KNeighborsClassifier()
knn.fit(X_train, y_train)

knn_pred = knn.predict(X_test)
```

[18]:
```python
lr = LogisticRegression(max_iter=200)
lr.fit(X_train, y_train)

lr_pred = lr.predict(X_test)
```

[19]:
```python
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)

dt_pred = dt.predict(X_test)
```

[20]:
```python
print("KNN Accuracy:", accuracy_score(y_test, knn_pred))
print("Logistic Regression Accuracy:", accuracy_score(y_test, lr_pred))
print("Decision Tree Accuracy:", accuracy_score(y_test, dt_pred))
```

```
KNN Accuracy: 1.0
Logistic Regression Accuracy: 1.0
Decision Tree Accuracy: 1.0
```

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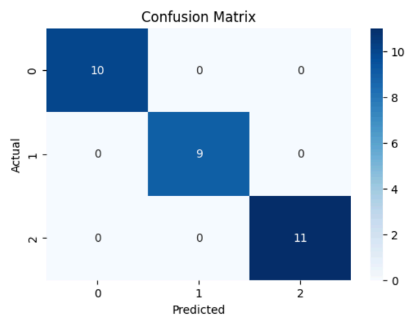
[20]:
```python
print("KNN Accuracy:", accuracy_score(y_test, knn_pred))
print("Logistic Regression Accuracy:", accuracy_score(y_test, lr_pred))
print("Decision Tree Accuracy:", accuracy_score(y_test, dt_pred))
```

```
KNN Accuracy: 1.0
Logistic Regression Accuracy: 1.0
Decision Tree Accuracy: 1.0
```

[21]:
```python
cm = confusion_matrix(y_test, knn_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



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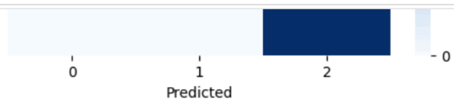

[22]:
```python
print(classification_report(y_test, knn_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

[23]:
```python
joblib.dump(knn, "iris_model.pkl")
```

[23]: ['iris_model.pkl']

[24]:
```python
model = joblib.load("iris_model.pkl")

sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)

print("Predicted Class:", prediction)
```

```
Predicted Class: [0]
```